

# LINUX



## Command: mkdir

---

To create a new directory use “mkdir”

```
/lukaj@DESKTOP-P2RPBKF:/$ sudo mkdir test  
[sudo] password for vlukaj:  
/lukaj@DESKTOP-P2RPBKF:/$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/$ ls  
bin  dev  home  lib    media  opt    root  sbin  srv  test  usr  
boot  etc  init  lib64  mnt    proc  run   snap  sys  tmp   var  
vlukaj@DESKTOP-P2RPBKF:/$ |
```

## Command: rmdir

---

To remove an empty directory use  
“rmdir”

```
vlukaj@DESKTOP-P2RPBKF:/$ sudo rmdir test
vlukaj@DESKTOP-P2RPBKF:/$ ls
bin  boot  dev  etc  home  init  lib  lib64  media  mnt  opt  proc  root  run  sbin  snap  srv  sys  tmp  usr  var
vlukaj@DESKTOP-P2RPBKF:/$ |
```

# Create a file: nano

```
vlukaj@DESKTOP-P2RPBKF:/test$ sudo nano test
vlukaj@DESKTOP-P2RPBKF:/test$ ls
test
vlukaj@DESKTOP-P2RPBKF:/test$ |
```



GNU nano 2.9.3 test

ciao a tutti

[ File 'test' is unwritable ]

^G Get Help ^O Write Out ^W Where Is ^K Cut Text ^J Justify ^C Cur Pos ^U Undo ^M Mark Text  
^X Exit ^R Read File ^B Replace ^N Uncut Text ^T To Spell ^\_ Go To Line ^E Redo ^D Copy Text

## Displaying a file

---

- Various ways to display a file in Unix
  - cat
  - less
  - head
  - tail

## Command: cat

---

- Dumps an entire file to standard output
- Good for displaying short, simple files

```
vlukaj@DESKTOP-P2RPBKF: /test$ cat test  
ciao a tutti  
vlukaj@DESKTOP-P2RPBKF: /test$ |
```

## Command: cat

---

- Dumps an entire file to standard output
- Good for displaying short, simple files

```
vlukaj@DESKTOP-P2RPBKF: /test$ cat test  
ciao a tutti  
vlukaj@DESKTOP-P2RPBKF: /test$ |
```

## Command: less

---

- “less” displays a file, allowing forward/backward movement within it
  - return scrolls forward one line, space one page
  - y scrolls back one line, b one page
- use “/” to search for a string
- Press **q** to quit



# Command: less

---

```
ciao a tutti  
test (END)
```

## Command: head

---

- “head” displays the top part of a file
- By default it shows the first 10 lines
- -n option allows you to change that
- “head -n50 file.txt” displays the first 50 lines of file.txt

## Command: tail

---

| Same as head, but shows the last lines

## File Commands

---

- Copying a file: `cp`
- Move or rename a file: `mv`
- Remove a file: `rm`

## File Commands: cp

---

To copy a file  
use “cp”

```
test.txt test2.txt  
vlukaj@DESKTOP-P2RPBKf:/mnt/c/Users/vluka/test$ cp test2.txt test.txt  
vlukaj@DESKTOP-P2RPBKf:/mnt/c/Users/vluka/test$ |
```

# File Commands: mv

- To move a file to a different location use “mv”

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ ls
test.txt  test2.txt
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ mkdir copy
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ ls
copy  test.txt  test2.txt
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ mv test.txt copy
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ ls
copy  test2.txt
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ cd copy/
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ ls
test.txt
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ |
```

## File Commands: mv

|mv can also be used to rename a file

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ ls  
copy  test2.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ cd copy/  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ ls  
test.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ mv test.txt newfile.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ ls  
newfile.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ |
```

## File Commands: rm

To remove a file use “rm”

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ ls  
copy  test2.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ cd copy/  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ ls  
test.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ |
```

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ mv test.txt newfile.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ ls  
newfile.txt  
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test/copy$ |
```



## File Permissions

---

- Each file in Unix/Linux has an associated permission level
- This allows the user to prevent others from reading/writing/executing their files or directories
- Use “ls -l *filename*” to find the permission level of that file

# File Permissions

---

A file belongs to a user and a group

There are 3 types of access:

- “r” means “read only” permission
- “w” means “write” permission
- “x” means “execute” permission
  - In case of directory, “x” grants permission to list directory contents

# Permission Levels

---

Directory:

- read (r) allows you to list the contents of the folder
- write (w) allows you to change the contents of the folder
- run(x) traverses a directory

# File Permission

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c$ ls -l
ls: cannot access 'hiberfil.sys': Permission denied
ls: cannot access 'pagefile.sys': Permission denied
ls: Recovery: Permission denied
ls: cannot access 'swapfile.sys': Permission denied
ls: 'System Volume Information': Permission denied
total 36
drwxrwxrwx 1 root root 4096 Feb 25 2020 '$Recycle.Bin'
dr-xr-xr-x 1 root root 4096 Nov 9 09:08 '$WINDOWS.BT'
dr-xr-xr-x 1 root root 4096 Oct 6 09:55 '$WinREAgent'
dr-xr-xr-x 1 root root 4096 Jul 14 2021 '$Windows.WS'
drwxrwxrwx 1 root root 4096 Dec 28 2019 DELL
lrwxrwxrwx 1 root root 12 Feb 7 2020 'Documents and Settings' -> /mnt/c/Users
drwxrwxrwx 1 root root 4096 Nov 28 2021 Downloads
drwxrwxrwx 1 root root 4096 Dec 28 2019 Drivers
-r-xr-xr-x 1 root root 834 Dec 17 2019 FIOD.manifest
drwxrwxrwx 1 root root 4096 Jul 4 2021 Favorites
drwxrwxrwx 1 root root 4096 Nov 9 10:51 Intel
drwxrwxrwx 1 root root 4096 Aug 3 2020 OneDriveTemp
drwxrwxrwx 1 root root 4096 Aug 5 2020 PerfLogs
dr-xr-xr-x 1 root root 4096 Nov 9 09:54 'Program Files'
dr-xr-xr-x 1 root root 4096 Sep 27 09:45 'Program Files (x86)'
drwxrwxrwx 1 root root 4096 Oct 18 09:38 ProgramData
d--x--x--x 1 root root 4096 Dec 28 2019 Recovery
d--x--x--x 1 root root 4096 Nov 9 12:33 'System Volume Information'
dr-xr-xr-x 1 root root 4096 Nov 28 2021 Users
dr-xr-xr-x 1 root root 4096 Oct 25 15:52 Windows
-r-xr-xr-x 1 root root 31827 Dec 28 2019 dell.sdr
-???????? ? ? ? ? ? hiberfil.sys
```

# File Permission

User(You)

The World

```
dr-xr-xr-x 1 root root 4096 Nov  9 09:08 '$WINDOWS~BT'  
dr-xr-xr-x 1 root root 4096 Oct  6 09:55 '$WinREAgent'  
dr-xr-xr-x 1 root root 4096 Jul 14 2021 '$Windows~WS'
```

Group

# File Permission: chmod

- If you own the file, you can change it's permissions with "chmod"
  - Syntax: chmod [**u**ser/**g**roup/**o**thers/**a**ll]+[permission] [file(s)]
  - Below we grant execute permission to all:

```
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ ls -l
total 0
drwxrwxrwx 1 root root 4096 Nov  9 13:47 copy
-rwxrwxrwx 1 root root  18 Nov  9 13:27 test2.txt
vlukaj@DESKTOP-P2RPBKF:/mnt/c/Users/vluka/test$ |
```

# File Permission: chmod

## | Linux File Permissions

 [blog.bytebytego.com](https://blog.bytebytego.com)

| Binary | Octal     | String Representation | Permissions            |
|--------|-----------|-----------------------|------------------------|
| 000    | 0 (0+0+0) | ---                   | No Permission          |
| 001    | 1 (0+0+1) | --x                   | Execute                |
| 010    | 2 (0+2+0) | -w-                   | Write                  |
| 011    | 3 (0+2+1) | -wx                   | Write + Execute        |
| 100    | 4 (4+0+0) | r--                   | Read                   |
| 101    | 5 (4+0+1) | r-x                   | Read + Execute         |
| 110    | 6 (4+2+0) | rw-                   | Read + Write           |
| 111    | 7 (4+2+1) | rwX                   | Read + Write + Execute |

| Owner |   |   | Group |   |   | Other |   |   |
|-------|---|---|-------|---|---|-------|---|---|
| r     | w | x | r     | w | - | r     | - | x |
| 4     |   |   | 2     |   |   | 1     |   |   |
| 7     |   |   | 6     |   |   | 5     |   |   |

## Special Characters

---

You can make use of jolly characters:

- '\*' group of characters including the '.' (non-initial)
- '?': a single character
- [...]: any character between those inside the brackets

The '~' character is interpreted as the user's home directory



# Commands

---

## **ls [options] [file]**

- ls
- ls -l
- display the contents of the root's var directory

## **cd [directory]**

- cd /tmp
- cd ciao/
- cd
- change the current directory to the root's usr directory

- mkdir test

- `mkdir /elements/test`
- create the **new test** directory in the user's home directory

- cp f1 f2 - r

- `cp f1 dir1`
- `cp f1 f2 f3 dir1`
- `cp *.doc *.tex` not working

# Commands

---

**rm [options] name...**

- `rm test`
- `rm ./test`
- `rm -r test/`
- `rm -r .*`

# Commands

---

**rm [options] name...**

- `rm test`
- `rm ./test`
- `rm -r test/`
- `rm -r .*`

# Commands

---

## **mv [options] source .... destination**

You cannot rename directories; to do this, use the mv command

- mv test test1
- mv \* /tmp

## **cat [options] [file]**

- concatenates files and sends them to standard output
- cat test test1
- cat test test1>test2

# Commands

## chmod [reference][operator][ mode ]file

The references are used to distinguish the users to whom the permissions apply i.e. they are list of letters that specifies whom to give permissions. The references are represented by one or more of the following letters:

| Reference | Class  | Description                                                            |
|-----------|--------|------------------------------------------------------------------------|
| u         | owner  | file's owner                                                           |
| g         | group  | users who are members of the file's group                              |
| o         | others | users who are neither the file's owner nor members of the file's group |
| a         | all    | All three of the above, same as ugo                                    |

# Commands

---

## chmod [reference][operator][ mode ]file

The operator is used to specify how the modes of a file should be adjusted. The following operators are accepted:

*Note : Putting blank space(s) around operator would make the command fail.*

| Operator | Description                                                                  |
|----------|------------------------------------------------------------------------------|
| +        | Adds the specified modes to the specified classes                            |
| -        | Removes the specified modes from the specified classes                       |
| =        | The modes specified are to be made the exact modes for the specified classes |

# Commands

---

## **chmod [reference][operator][ mode ]file**

The modes indicate which permissions are to be granted or removed from the specified classes. There are three basic modes which correspond to the basic permissions:

```
r      Permission to read the file.  
w      Permission to write (or delete) the file.  
x      Permission to execute the file, or, in  
        the case of a directory, search it.
```



# Commands

---

## **chmod [reference][operator][ mode ]file**

- `chmod o-r my_file`
- `chmod a+w my_file`

Ex:

- `touch my_file`
- `chmod -r my_file`
- `ls -l my_file`
- `cat my_file`
- `chmod +r my_file`
- `chmod -r my_file`
- `ls > my_file`
- `mv my_file you_file`
- `ls -l you_file`
- `rm you_file`

# Commands

---

## Permissions in directories:

- `mkdir test`
- `touch test/first`
- `touch test/second`
- `chmod -r test`
- `ls test`
- `chmod +r test`
- `chmod -w test`
- `rm test/first`
- `cp test/first test/third`
- `mv test/first test/third`
- `chmod +w test`
- `mkdir test/tmp`
- `touch test/tmp/example`
- `chmod -x test`
- `cd test`
- `cat test/tmp/example`
- `touch test/tmp/example2`